

Group Homework

Group 4

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I. Fraud Detection Confusion Matrix

1. If banks use 0.4 as the cutoff to classify the predicted transaction to be fraud.

Transaction ID	Transaction Amount	Fraud Indicator (Actual)	Predicted Probability	Fraud Indicator (Cutoff = 0.4)
5	\$30	F	0.05	F
4	\$50	T	0.14	F
8	\$100	T	0.32	F
7	\$700	F	0.48	T
1	\$770	F	0.55	T
6	\$990	T	0.61	T
3	\$1200	F	0.72	T
2	\$1300	T	0.82	T

Confusion Matrix:

	Predicted Positive	Predicted Negative
Actual Positive	2 (TP)	2 (FN)
Actual Negative	3 (FP)	1 (TN)

Calculation of True Positive Rate and False Positive Rate:

True Positive Rate = $TP / (TP + FN) = TP / \text{Actual Positive}$

$$= 2 / (2+2)$$

$$= 2 / 4$$

$$= 1 / 2$$

$$= 0.5$$

True Positive Rate = 50%

False Positive Rate = FP / (FP + TN) = FP / Actual Negative

$$= 3 / (3+1)$$

$$= 3/4$$

$$= 0.75$$

False Positive Rate = 75%

2. If banks use 0.6 as the cutoff to classify the predicted transaction to be fraud.

Transaction ID	Transaction Amount	Fraud Indicator (Actual)	Predicted Probability	Fraud Indicator (Cutoff = 0.6)
5	\$30	F	0.05	F
4	\$50	T	0.14	F
8	\$100	T	0.32	F
7	\$700	F	0.48	F
1	\$770	F	0.55	F
6	\$990	T	0.61	T
3	\$1200	F	0.72	T
2	\$1300	T	0.82	T

Confusion Matrix:

	Predicted Positive	Predicted Negative
--	--------------------	--------------------

Actual Positive	2 (TP)	2 (FN)
Actual Negative	1 (FP)	3 (TN)

Calculation of True Positive Rate and False Positive Rate:

True Positive Rate = $TP / (TP + FN) = TP / \text{Actual Positive}$

$$= 2 / (2+2)$$

$$= 2 / 4$$

$$= 1 / 2$$

$$= 0.5$$

True Positive Rate = 50%

False Positive Rate = $FP / (FP + TN) = FP / \text{Actual Negative}$

$$= 1 / (1+3)$$

$$= 1/4$$

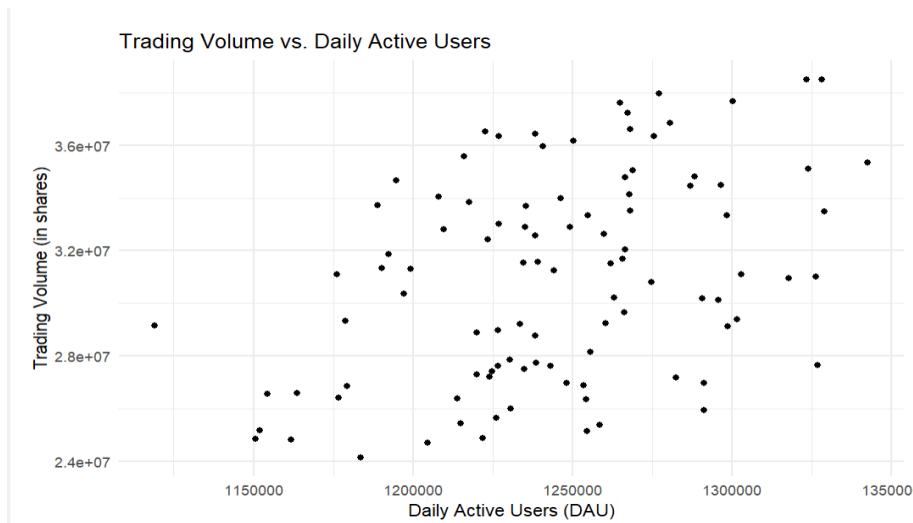
$$= 0.25$$

False Positive Rate = 25%

3. When the bank sets the cutoff at 0.4, more transactions are marked as fraudulent, which helps catch more real frauds but also increases false alarms. At a cutoff of 0.6, the bank misses some frauds but reduces false positives. In this case, both cutoffs have the same true positive rate (TPR) of 0.5, but the false positive rate (FPR) drops from 0.75 to 0.25 at the higher cutoff. This means the higher cutoff performs better by keeping accuracy while reducing false alerts. The difference between the two cutoffs can be visualized using an ROC curve.

II. Daily active user and Trading volume on Robinhood (5 points)

1. Create a scatter plot to visualize the relationship between daily active users (x-axis) and trading volume (y-axis)



2. Perform a linear regression with trading volume as the outcome variable and daily active users as the independent variables.

```

{r}

model <- lm(`Trading.Volume..Shares.Traded.` ~ `Daily.Active.Users..DAU.`, data = robinhood_data)
summary(model)

```

```

Call:
lm(formula = Trading.Volume..Shares.Traded. ~ Daily.Active.Users..DAU.,
    data = robinhood_data)

Residuals:
    Min       1Q   Median       3Q      Max
-6638607 -2830705  146840  2686288  6310680

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) -1.244e+07  9.870e+06  -1.26   0.211
Daily.Active.Users..DAU.  3.487e+01  7.924e+00   4.40 2.75e-05 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 3580000 on 98 degrees of freedom
Multiple R-squared:  0.165,    Adjusted R-squared:  0.1565
F-statistic: 19.36 on 1 and 98 DF,  p-value: 2.752e-05

```

3. What is the slope of the regression line? Interpret the slope in the context of the problem.
 - The slope is the Estimate value for your predictor variable (Daily.Active.Users..DAU.). Slope here is 3.487e+01, which is scientific notation for 34.87.
 - The interpretation is: For each one-unit increase in the x-variable, the y-variable is predicted to change by the slope.
 - In this context: For every additional daily active user, the trading volume is predicted to increase by 34.87 shares.

4. Use the model to predict the trading volume if the number of daily active users is 1,500,000.

```
#Predicting trading volume if daily active users is 1500000
```{r}
1. Create a new data frame for the value we want to predict.
new_user_data <- data.frame(`Daily.Active.Users..DAU.` = 1500000)

2. Use the predict() function
predicted_volume <- predict(model, newdata = new_user_data)

3. Print the result
print(paste("Predicted trading volume for 1,500,000 DAU:", predicted_volume))
```

[1] "Predicted trading volume for 1,500,000 DAU: 39863304.715444"
```

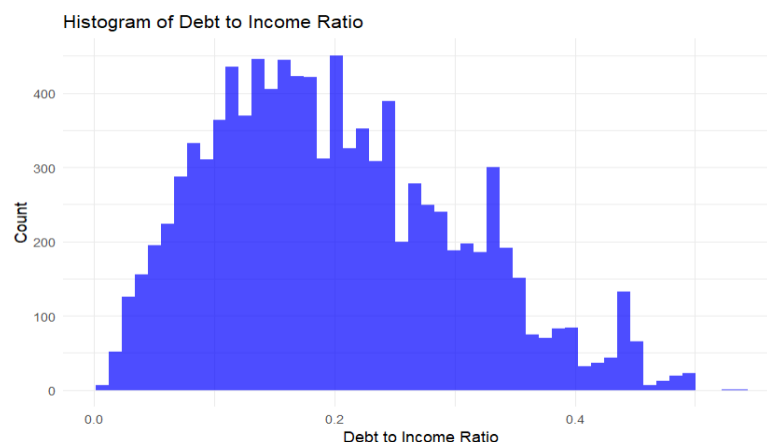
III. Predict consumer loan defaults (10 points)

1. Create a new variable called 'default dummy,' where 1 indicates a loan default and 0 indicates no default.

```
95 # Create 'default.dummy'
96 ```{r}
97 loan_data$default.dummy <- ifelse(loan_data$default == "Defaulted", 1, 0)
98 print(table(loan_data$default.dummy))
99 ```

      0      1
8377 1623
```

2. You also want to create debt to income ratio based on the data. Debt to income ratio can be defined as the borrower's loan amount to annual income ratio. You can name the new variable as **debt_to_income**. Draw a histogram to see how the distribution looks like for this new variable.



3. Estimate a logistic regression model with debt-to-income ratio as the predictor. Report your estimates for the intercept and the coefficient for debt to income. What does the coefficient tell you on the relationship between debt-to-income ratio and default probability?

Based on your R output, here are the estimates from your logistic regression model:

- **Intercept Estimate: -2.12174**
- **Coefficient for debt_to_income: 2.31572**

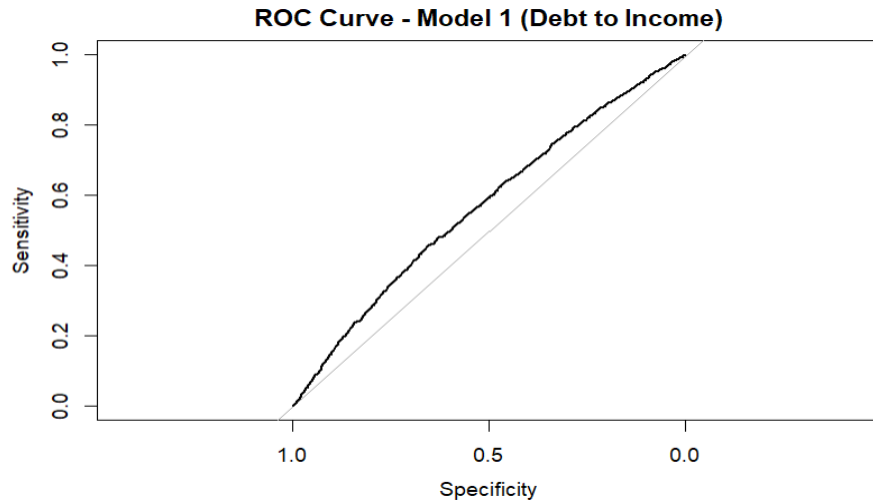
Interpretation of the Coefficient

The coefficient for debt_to_income is **positive (2.31572)**.

This tells you that there is a **positive relationship** between the debt-to-income ratio and the probability of defaulting. In simple terms:

As a borrower's debt-to-income ratio **increases**, their predicted probability of defaulting on the loan also **increases**.

4. Plot an ROC curve corresponding to the estimated model. What is the area under the curve? Is the area below your ROC curve more significant than 0.5? Report on the area.



Area under the curve (AUC): 0.5697

Yes. An AUC of **0.5697** is greater than 0.5.

Interpretation: An AUC of 0.5 represents a model with no predictive power (equivalent to random guessing). Since your AUC is **0.5697**, model's ability to distinguish between a defaulting loan and a non-defaulting loan is slightly better than random chance.

5. Suppose you develop a policy that rejects all the loan applicants whose predicted probability of default is greater than 20%.

```
[1] "--- Part 5: Confusion Matrix (Model 1 @ 0.20) ---"
      Predicted
Actual    0     1
      0 7220 1157
      1 1290  333

# 5.1) Proportion correctly predicted to default (True Positive Rate)
{r}
tpr1 <- conf_matrix1[2, 2] / sum(conf_matrix1[2, ])
print(paste("5.1: Proportion correctly predicted to default (TPR):", tpr1))

[1] "5.1: Proportion correctly predicted to default (TPR): 0.205175600739372"

# 5.2) Proportion mistakenly predicted to default (False Positive Rate)
{r}
fpr1 <- conf_matrix1[1, 2] / sum(conf_matrix1[1, ])
print(paste("5.2: Proportion mistakenly predicted to default (FPR):", fpr1))

[1] "5.2: Proportion mistakenly predicted to default (FPR): 0.138116270741315"
```

1) What is the proportion of consumers you correctly predicted to default (The denominator should be the total number of people who defaulted.)

Proportion correctly predicted to default: 0.205175 or 20.52%

- This is the **True Positive Rate (TPR)**. R output shows it was calculated by dividing the 333 people you correctly predicted would default by the 1,623 people who *actually* defaulted ($333 / (1290 + 333)$).

2) What is the proportion of consumers you mistakenly predicted to default? (The denominator should be the total number of people who do not default.)

Proportion mistakenly predicted to default: 0.138116... or 13.81%

- This is the **False Positive Rate (FPR)**. R output shows it was calculated by dividing the 1,157 people you *mistakenly* predicted would default by the 8,377 people who *did not* default ($1157 / (7220 + 1157)$).

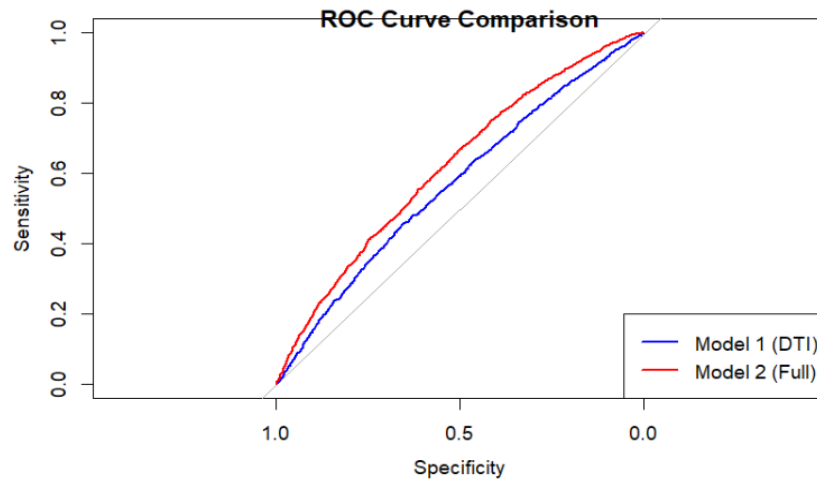
Suppose now you include additional variables in your model. You must run a logistic regression with FICO, debt to income ratio, and revolving credit balance.

6. Run a logistic regression using the explanatory variables listed above. Report your estimates for the intercept and the coefficients for all variables.

Based on your R output for Model 2, here are the estimated coefficients:

- **Intercept:** 7.397e+00 (which is 7.397)
- **Coefficient for fico:** -1.373e-02 (which is -0.01373)
- **Coefficient for debt_to_income:** 2.377e+00 (which is 2.377)
- **Coefficient for revol_bal:** 8.246e-07 (which is 0.0000008246)

7. Plot the ROC curve. Is the area below your new ROC curve greater than that in part 3? Report on the area. Interpret the differences.
Based on your R output and plot, here is the analysis:



- **Yes.** The new model's AUC is **0.6204** (rounded), which is greater than the first model's AUC of **0.5697**.
- **Report on the area:**
 - **Model 1 (DTI only) AUC:** 0.5697
 - **Model 2 (Full model) AUC:** 0.6204

Interpretation

The AUC measures the model's ability to correctly distinguish between a positive case (default) and a negative case (no default). A higher AUC means a better model.

The new model (Model 2) is **a better predictive model** than the first model. By adding the FICO score and revolving credit balance (revol_bal) as predictors, the model's ability to separate defaulters from non-defaulters has **improved**.

Visually, your "ROC Curve Comparison" plot shows this: the **red line (Model 2)** is slightly higher and more "bowed" toward the top-left corner than the **blue line (Model 1)**, confirming it has a larger area under the curve.

8. Suppose you develop a policy that rejects all the loan applicants whose predicted probability of default is greater than 10%.

```
[1] "--- Part 8: Confusion Matrix (Model 2 @ 0.10) ---"
      Predicted
Actual    0     1
      0 1272 7105
      1   108 1515

# 8.1) Proportion correctly predicted to default (True Positive Rate)
```{r}
tpr2 <- conf_matrix2[2, 2] / sum(conf_matrix2[2,])
print(paste("8.1) Proportion correctly predicted to default (TPR):", tpr2))
```

[1] "8.1) Proportion correctly predicted to default (TPR): 0.933456561922366"

# 8.2) Proportion mistakenly predicted to default (False Positive Rate)
```{r}
fpr2 <- conf_matrix2[1, 2] / sum(conf_matrix2[1,])
print(paste("8.2) Proportion mistakenly predicted to default (FPR):", fpr2))
```

[1] "8.2) Proportion mistakenly predicted to default (FPR): 0.848155664318969"
```

1) What is the proportion of consumers you correctly predicted to default (The denominator should be the total number of people who defaulted.)

Proportion correctly predicted to default:

- **0.933456** or **93.35%**
- This is the **True Positive Rate (TPR)**. Your R output shows it was calculated from the confusion matrix by dividing the 1,515 people you correctly predicted would default by the 1,623 people who *actually* defaulted ($1515 / (108 + 1515)$).

2) What is the proportion of consumers you mistakenly predicted to default? (The denominator should be the total number of people who do not default.)

Proportion mistakenly predicted to default:

- **0.848155** or **84.82%**
- This is the **False Positive Rate (FPR)**. Your R output shows it was calculated by dividing the 7,105 people you *mistakenly* predicted would default by the 8,377 people who *did not* default ($7105 / (1272 + 7105)$).